Asim Munawar

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PROFESSIONAL SUMMARY

Senior AI Research Leader with 15+ years of experience leading applied and foundational AI research, and designing, scaling, and deploying **LLM-based reasoning and agentic systems** in production. Proven leader of large, cross-functional AI teams (up to 50+ researchers) delivering **measurable improvements in model reasoning, system reliability, and enterprise adoption**. Deep expertise in **LLM training, synthetic data, reinforcement learning, evaluation frameworks, and GPU-accelerated AI systems**, with a strong track record of translating research into real-world impact.

CORE SKILLS

LLMs - Reasoning & Agentic Systems
Large Language Models (LLMs), multi-step reasoning, planning, agentic workflows, tool-augmented generation, grounded reasoning

Model Training & Optimization
Supervised fine-tuning (SFT), reinforcement learning, synthetic data generation, evaluation-driven optimization

Systems & Deployment
Distributed training, scalable AI systems, GPU-accelerated workflows, research-to-production transfer, model optimization

Leadership & Collaboration
Technical leadership, cross-functional collaboration, hiring & mentoring, roadmap definition, stakeholder alignment

WORK EXPERIENCE

IBM Research – New York
2012 - Present

Technical Lead – LLM Reasoning & Agentic Systems
Jan 2023 – Present

- Led the design and deployment of **production-grade LLM reasoning and agentic workflows**, improving reasoning benchmark performance by **20%+** and enterprise task success by **30%+.**
- Managed and technically guided a **20+ person cross-functional research team** spanning LLM training, evaluation, and systems integration.
- Drove **synthetic data generation, supervised fine-tuning, and reinforcement learning pipelines** to improve robustness and generalization of large-scale models.
- Delivered **multi-turn grounded reasoning systems** adopted across internal production pipelines and enterprise AI platforms.
- Partnered with systems and infrastructure teams to align model design with **GPU-efficient training and inference constraints**.

Program Director – Neuro-Symbolic AI
Jul 2019 – May 2025

- Co-designed and led a **global research program** spanning **50+ researchers across 7 international teams**.
- Defined technical strategy for **hybrid neural-symbolic reasoning**, reinforcement learning, and real-world AI decision systems.
- Built and scaled an **external research community of 6,000+ members**, accelerating collaboration, adoption, and knowledge transfer.
- Influenced long-term research roadmaps and product-adjacent initiatives across multiple IBM organizations.

Research Manager – Embodied Learning
Jan 2018 – Nov 2019

- Managed a **12+ member research team** working across NLP, robotics, computer vision, and acoustics.
- Owned hiring, mentoring, performance evaluation, and enterprise-impact research delivery.
- Acted as a **bridge between research and product teams**, ensuring alignment with business needs.

Senior Researcher
Apr 2012 – Jan 2018

- Conducted applied research in **robotics, computer vision, deep learning, and neuromorphic computing**.
- Built simulation platforms and emulators for intelligent systems and AI-driven automation.
- Founded an applied AI research team and co-led internal innovation initiatives.

University of Michigan at Flint
Oct 2025 – Present
Jack W. Thompson, M.D. Distinguished Visiting Professor

- Keynote Speaker & Panelist at various events
- Lecturer & Master Class Instructor for students, faculty, and community audiences
- Research & Industry Collaboration Lead (AI/cybersecurity consulting, grants, and joint initiatives)

Early Career Roles (Selected)

- **Quality Assurance Engineer, Digital Works** — Japan (2010–2012)
- **Campus Ambassador, Sun Microsystems** (2008–2010)
- **Web / Android Developer, Self-Employed** (2006–2012)
- **Design Engineer, CARE** — DSP / FPGA Systems (2003–2005)

Advisory & Board Roles

- **Executive Board Member**, Centaur AI (2023–Present)
  Advising on generative AI strategy and responsible deployment.
- **SIAB Board Member**, National Center for AI, NCAI Pakistan (2021–Present)
  Guiding national AI research direction and performance outcomes.
- **Editorial Advisory Board**, Journal of Aerospace & Security Studies, JASS (2025–Present)

EDUCATION

Postdoctoral Researcher, Hokkaido University, Japan
2011 – 2012 — Academic cloud platforms for large-scale research

Ph.D., Computer Science, Hokkaido University, Japan
2008 – 2011 — Parallel evolutionary algorithms

M.Sc., Computer Science, Hokkaido University, Japan
2006 – 2008 — Grade A+

B.E., Computer Systems Engineering, NUST, Pakistan
1999 – 2003 — CGPA: 3.875 / 4.0

RESEARCH IMPACT & RECOGNITION

- 20+ US patents | 80+ peer-reviewed publications | 3 Best paper awards
- Regular invited speaker at international conferences, workshops, and industry events
- ***Selected Corporate Awards***
  - IBM A-level Award – Synthetic Data Generation Framework (2025)
  - IBM O-level Award – Granite 3.0 (2024)
  - IBM A-level Award – API-augmented LLMs (2024)

CERTIFICATIONS & ADDITIONAL INFORMATION

- IBM Certified Manager
- Sun Certified Associate, Java Platform (SE)
- Languages: English, Japanese, Urdu/Hindi, Punjabi
- U.S. Permanent Resident